UNDERGRADUATE & RESIDENCY CLINICAL SKILLS TEACHING POSTER · CARDIORENAL MODULE

Acute Decompensated Heart Failure: Pathophysiology, Pharmacology & OSCE Guide

Integrate Frank-Starling hemodynamics, bedside Forrester perfusion-congestion profiling, diuretic pharmacology, and structured cardiovascular OSCE examination.

1. ANATOMY, PHYSIOLOGY & MECHANISM INTEGRATION

1 **Neurohormonal Activation:**

Falling cardiac output triggers sympathetic surge + RAAS + AVP release, elevating preload and systemic vascular resistance.

2 **Venous Congestion & Starling Forces:**

LVEDP > 18 mmHg transmits backward to pulmonary capillaries, driving interstitial and alveolar transudation.

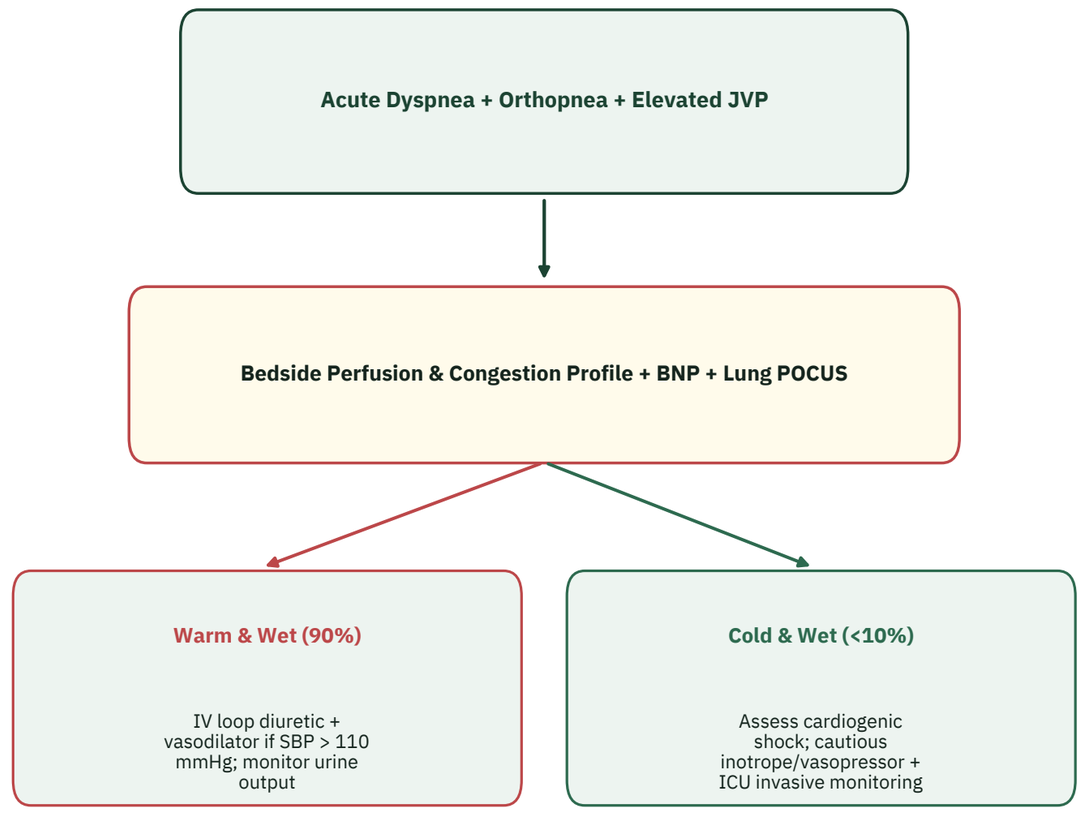
3 **Cardiorenal Venous Hypertension:**

Elevated central venous pressure reduces transrenal perfusion gradient, lowering eGFR independent of forward output.

4 **Frank-Starling Plateau:**

Overstretched sarcomeres operate on flat ventricular curve where vasodilator afterload reduction restores stroke volume.

2. STEPWISE DIAGNOSTIC DECISION ALGORITHM



3. HIGH-YIELD HEART FAILURE PHARMACOLOGY MATRIX

Drug Class	Site & Mechanism	Clinical Role	Key Adverse Effect / Monitoring
Loop Diuretics (Furosemide)	Inhibits thick ascending limb NKCC2 cotransporter	Rapid decongestion in volume overload	Hypokalemia, hypomagnesemia, ototoxicity, Cr rise
Vasodilators (IV Nitroglycerin)	Endothelial NO -> cGMP venodilation & afterload drop	Hypertensive acute pulmonary edema	Hypotension; contraindicated with PDE5 inhibitors
ARNI (Sacubitril/Valsartan)	Neprilysin + AT1 blockade (raises natriuretic peptides)	GDMT mortality reduction post-stabilization	36h washout from ACEi mandatory (angioedema risk)
SGLT2 Inhibitors (Empagliflozin)	Proximal tubule natriuresis + myocardial energetics	All-EF HF hospitalization & CV death reduction	Euglycemic ketoacidosis; genital mycotic infection

4. OSCE / OSPE STRUCTURED CLINICAL STATION CHECKLIST

Step 1 — Position patient at 45 degrees; inspect respiratory rate, accessory muscle use, and peripheral cyanosis.

Step 2 — Measure Jugular Venous Pressure (JVP) height above sternal angle with tangential illumination.

Step 3 — Palpate apex beat (5th ICS MCL vs laterally displaced/thrusting) and parasternal RV heave.

Step 4 — Auscultate S3 gallop (bell at apex in left lateral decubitus) and bibasilar inspiratory crackles.

Step 5 — Assess dependent sacral/pretibial pitting edema and tender congestive hepatomegaly.

CRITICAL SAFETY RED FLAGS

- **SBP < 90 mmHg with cool mottled extremities, oliguria (<0.5 mL/kg/h), or rising lactate (cardiogenic shock).**
- **SpO2 < 90% refractory to supplemental O2 requiring immediate CPAP/BiPAP ventilatory support.**
- **Never initiate beta-blockers acutely during uncompensated pulmonary edema or low-output shock.**
- **STEMI / acute mechanical valve or papillary muscle rupture on urgent 12-lead ECG & echo.**